Why did Medicare ACOs reset their benchmarks early? A pre-registered analysis of early renewal in the Medicare Shared Savings Program

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Keywords: accountable care organizations; Medicare Shared Savings Program; benchmarks; alternative payment models; preregistration; difference-in-differences; Bayesian analysis
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Abstract

Importance: Medicare Shared Savings Program (MSSP) benchmarks are reset at each new agreement from an accountable care organization's (ACO's) recent spending, adjusted toward its region. ACOs could end an agreement early to reset sooner, but who used that option, and why, has not been studied.

Objective: To test, under a preregistered plan, whether ACOs that renewed early stood to gain from the regional part of a benchmark reset, and whether their spending changed afterward.

Design, Setting, and Participants: Preregistered retrospective cohort study of the 334 MSSP ACOs whose agreements began July 1, 2019 or January 1, 2020, using CMS public use files (2014-2026). The registration (OSF) preceded all reading of outcome data, enforced by a hash-based lock in the analysis code. The reset benchmark was rebuilt from 42 CFR 425.601 and validated against benchmarks CMS published for 101 other ACOs before use.

Exposures: Early renewal: ending the agreement to begin a new one on January 1, 2022 (29 ACOs), compared with staying (277 ACOs).

Main Outcomes and Measures: Primary, the regional pull (case-mix-adjusted regional spending relative to the current benchmark). Secondary, the own-spending component of the reset, savings rates, a Bayesian logistic decision model with measurement error, and log per-capita spending in 2022-2024 (difference-in-differences with an equivalence condition on pre-trends). Decision rules were fixed in advance; significance was by permutation and direct simulation.

Results: The rebuilt benchmark tracked CMS's (Spearman ρ = 0.94; median absolute error 1.6%). Early renewers' regions spent 3.4 percentage points of benchmark more than stayers' (95% credible interval [CrI], 1.7 to 5.4; $P < .001$). Early renewers were also beating their benchmarks by more (own-spending component −2.1 points; 95% CrI, −3.8 to −0.5), so the total expected gain from a reset did not differ (−0.5 points; 95% CrI, −2.0 to 0.8). Among 206 BASIC-track ACOs, early renewal was associated with a higher regional pull and with higher (two-sided) BASIC levels. Spending growth after renewal did not differ detectably (−0.4%; 95% CrI, −4.0% to 2.5%; $P = .78$), but the preregistered equivalence condition did not hold, so this is not read as an effect.

Conclusions and Relevance: ACOs that reset early were those whose regions spent more relative to their benchmarks and those at higher BASIC levels, for whom renewal opened the 75% sharing rate. A reset was not, on average, a route to a higher benchmark. These associations describe the 2022 rules; benchmarks set from 2024 credit prior savings, which may change who renews early.

Introduction

In the Medicare Shared Savings Program (MSSP), an accountable care organization (ACO) shares in savings when the spending of its assigned beneficiaries falls below a benchmark.[1] At the start of each agreement period the benchmark is reset from the ACO's own spending in the 3 preceding years and adjusted part of the way toward the spending of its region.[2] Evaluations have found modest savings from the program, growing with tenure and larger in physician-group ACOs,[3–5] alongside concern that selection, of clinicians, beneficiaries or whole ACOs, shapes both the estimates and who participates.[6–8]

Benchmarks are central to that selection. Rebasing to an ACO's own recent spending weakens its incentive to save, and under earlier weights it rewarded higher spending just before a new agreement.[9] ACOs' spending varies widely relative to their regions even after risk adjustment, so moving benchmarks toward regional averages shifts them unevenly.[10] After regional spending entered benchmarks in 2017, participation shifted toward providers whose spending was low relative to their region,[11] ACOs whose effective benchmarks fell became more likely to leave,[12] and bonuses rose for ACOs across the efficiency spectrum.[13] Survival in the program has been associated with earning bonuses, higher benchmarks

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and one-sided risk.[14,15] The 2019 Pathways to Success rules, which shortened the time ACOs could stay in one-sided risk, were followed by more exits than entries,[16] by ACO leaders reassessing potential savings against losses,[17] and by departures with projected costs to Medicare.[18,19]

The same rules allowed an ACO to end its agreement early and begin a new one, which resets the benchmark early.[1] For an ACO in the BASIC track, renewal was also the only way into the ENHANCED track before its agreement ended, because the 2018 rule bars moving from BASIC to ENHANCED within an agreement period;[20] ENHANCED shares 75% of savings, against at most 50% in BASIC.[1] Of the 334 ACOs whose agreements began July 1, 2019 or January 1, 2020, 29 ended them early and began new agreements on January 1, 2022.[21] Who used the option, and why, bears on how benchmark rules sort ACOs, but to our knowledge it has not been studied.

Two motives predict early renewal. Under a **reset motive**, an ACO renews because its new benchmark would be higher. Under a **sharing-rate motive**, a BASIC-track ACO renews to reach ENHANCED. The savings rate cannot separate them: an ACO beating its benchmark would lose on the own-spending part of a reset, yet may gain on the regional part if its region spends more. This study therefore took as its primary test the **regional pull**, a measure of how far an ACO's case-mix-adjusted regional spending sat above its current benchmark, which contains no term for the ACO's own benchmark-year spending. It asked whether early renewers stood to gain more from the regional part of a reset than ACOs that stayed, which characteristics predicted early renewal, and whether early renewers' spending changed afterward. The analysis plan, hypotheses, decision rules and code were registered before any outcome value was read.[22]

Methods

Study design and preregistration

This is a preregistered retrospective cohort study of ACO-level public data, reported following the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guideline[23] (checklist in Supplement 1). The analysis plan and the code commit that implements it were registered on OSF Registries on October 26, 2026, 13:22 UTC, with the plan's SHA-256 hash recorded.[22,24] Before registration only the enrollment files, the CMS data dictionaries, the regulation text and the column names (not the values) of the outcome files were used. The registry's own timestamp, rather than the repository's commit dates, establishes that order, and the code enforces it: every data file is hashed before it is read, and any file whose hash matches a performance-year results file or a county spending file could not be opened until the registration record existed. The pipeline was written and tested on synthetic data with planted effects before registration. Analyses ran in stages that each refused to start

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until the outputs they depend on existed, so the power analysis was committed before any confirmatory estimate and every results file before its estimates were read.

Data sources

Four CMS public use files were used, all aggregate at the ACO or county level:[21,25–27] MSSP ACO enrollment files (one per year, 2014-2026), performance-year financial and quality results (2013-2024), county-level expenditure and risk-score data on assignable beneficiaries, and counts of ACO-assigned beneficiaries by county. Forty files were downloaded on October 22, 2026, and each file's size, SHA-256 hash and URL was recorded; an archived copy of the snapshot is on OSF.[28] Cells with counts of 1 to 10 are suppressed in these files.[29]

Population and groups

The base population was the 334 ACOs whose performance year (PY) 2021 enrollment record showed an agreement start of July 1, 2019 (189 ACOs) or January 1, 2020 (145 ACOs). Both cohorts' agreements end December 31, 2024, so an ACO from either cohort with a January 1, 2022 agreement in the PY2022 enrollment file had renewed early.[1] **Early renewers** (29 ACOs) were compared with **stayers** (277 ACOs), those in the same agreement in PY2022. The 28 ACOs absent from the PY2022 file (left after PY2021) were described but entered no registered test (Figure 1).

Variables

Expected gain from a reset. CMS does not publish what an early renewer's benchmark would have been had it not renewed, nor a stayer's had it reset. Following 42 CFR 425.601 as in force on January 1, 2022,[2] the registered plan rebuilt it in 6 steps for each of the 4 enrollment types (end-stage renal disease [ESRD], disabled, aged/dual eligible, aged/non-dual eligible): (1) the equally weighted mean of the ACO's per-capita spending in PY2019-PY2021, restated to PY2021 case mix and trended to 2021 dollars with the national-regional growth blend; (2) risk-standardized county spending averaged with weights equal to the ACO's assigned person-years by county in 2021 and adjusted to the ACO's case mix;[30] (3) the regional adjustment, $w \times (\text{regional} - \text{rebased})$, with the phase-in weight w and the cap of 425.601; (4) the approximate new benchmark, combining types by the ACO's PY2021 person-year shares; and (5) the expected gain, $100\times$ (new $-$ current) / current, where the current benchmark is the PY2021 updated per-capita benchmark. The gain splits into an **own-spending component** (rebased spending relative to the current benchmark, which is close to a restatement of the savings rate) and a **regional component** (the adjustment). Writing the adjustment out shows that, before capping,

$$\text{regional component} = w \times (\text{pull} - \text{own}),$$

so the regional component is also tied to past savings. Step (6) is the primary exposure measure, the **regional pull**,

$$\text{regional pull} = 100 \times \frac{\sum_e s_e R_e - B}{B},$$

where R_e is the case-mix-adjusted regional spending for enrollment type e, s_e the ACO's share of person-years of that type, and B its current per-capita benchmark. All 4 are in percentage points of the current benchmark.

Validation before use. Several steps can only be approximated from public files. The plan therefore required the rebuilt benchmark to be validated against the 2022 benchmarks CMS published for regular renewers (ACOs whose scheduled agreements ended in 2021) before any expected-gain variable was estimated, with a pass rule of a Spearman correlation of at least 0.80 and a median absolute percentage error of at most 5%, and a calibration step if the slope of actual on rebuilt fell outside 0.9 to 1.1. Early renewers were excluded from the validation sample and the 2022 results file was filtered as it was loaded, so validation could not reveal their actual reset. A second check compared the own-spending part with CMS's own benchmark-year spending for the same ACOs; its error SD set the measurement-error SD used in the decision model.

Other variables. From the results files: the savings rate (CMS-reported savings as a share of the benchmark) in PY2020 and PY2021, assigned beneficiaries, and log per-capita spending, PY2020-PY2024. From the enrollment files: cohort, PY2021 track and level, revenue category, assignment method and agreement number.

Suppressed values. Hidden cells needed for the expected gain were filled by registered rules in order: exact recovery of a single hidden count from totals, the mean over years in which a value was shown, and a regional ratio. ACOs still missing a variable were excluded from that analysis and counted; the plan required reporting as a limitation any exclusion above 10% of either group.

Hypotheses

The registered hypotheses were: **H2b (primary)**, early renewers had a larger regional pull than stayers; **H2a (secondary)**, a larger own-spending component (close to H1 restated, so not counted as separate evidence); **H1 (descriptive)**, a difference in PY2020 savings rate, with no motive attributed to its sign; **H3**, among BASIC-track ACOs, the probability of renewing early rises with the regional pull, the own-spending component and a higher BASIC level; and **H4**, early renewal changed per-capita spending in

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PY2022-PY2024 (2-sided). H1 to H3 describe who chose to reset and are associations. H4 concerns a bundled package (a new benchmark, for almost all a move to ENHANCED, and new benchmark years) whose parts cannot be separated.

Statistical analysis

Group comparisons (H1, H2a, H2b). The estimate was the difference in means, early renewers minus stayers, within each cohort, averaged with weights proportional to the early renewers in each cohort. Its posterior was a Bayesian bootstrap[31] (10,000 draws of Dirichlet(1, …, 1) weights within each cohort-by-group cell), reported as the posterior mean and 95% equal-tailed credible interval (CrI). Significance was by stratified permutation: labels were shuffled within cohort 10,000 times, and the 2-sided *P* value was $(1 + \#\{|\hat{\theta}_{\text{perm}}| \geq |\hat{\theta}_{\text{obs}}|\})/(1 + 10{,}000)$.[32]

Decision model (H3). A Bayesian logistic regression of early renewal among PY2021 BASIC-track ACOs on the 3 core predictors and 5 adjustment factors (cohort, log assigned beneficiaries, high revenue, prospective assignment, later agreement) was fit in PyMC[33] with the No-U-Turn sampler[34] (4 chains of 2,000 draws). Continuous predictors were scaled by 2 SD,[35] so each coefficient is a log-odds difference per 2 SD; coefficients had Normal(0, 1) priors. The own-spending component entered as a latent true value observed with normal error, with the error SD fixed by the validation step. Convergence required $\hat{R} \leq 1.01$, bulk and tail effective sample sizes of at least 400 and no divergences,[36] with one registered rerun; a Firth penalized regression[37,38] would have decided H3 had the model not converged. Because shuffling labels in a multi-predictor model tests the joint null, each core predictor was tested by direct simulation under the reduced model: the model without that predictor was fit, 10,000 outcome vectors were simulated from its posterior, the full model was refit to each, and the observed coefficient was compared with that null distribution. A Student-t(3, 0, 2.5) prior fit[39] was the registered sensitivity analysis.

Spending after renewal (H4). For each ACO, Δ was the mean log per-capita spending over its included years in PY2022-PY2024 minus the mean in PY2020-PY2021, and H4 applied the group-comparison estimator to Δ, reported as $100 \times (e^{\text{estimate}} - 1)$, the percent difference in spending growth. The comparison group was ACOs not yet renewed: a stayer's years after it renewed into a 2023 or 2024 agreement were excluded, following the not-yet-treated design of Callaway and Sant'Anna.[40] With one pre-period change (PY2020 to PY2021, both affected by COVID-19) and about 29 treated ACOs, a nonsignificant pre-trend test is weak evidence of parallel trends,[41] so the plan replaced it with an equivalence condition:[42] H4 would be read as an effect only if the larger absolute bound of the pre-trend CrI, scaled by the mean distance between period midpoints, was smaller than the absolute H4 estimate. A trend-adjusted estimate[43] and risk-adjusted, balanced-panel, propensity-weighted[44] and single-cohort estimates were reported beside it.

Power. Minimum detectable effects (MDEs) at 80% power and 2-sided α = .05 were computed after reg-

istration from spreads in years or groups not used in the tests, and committed before any confirmatory estimate. For H3, detection was simulated at a log-odds difference of 0.5.

Decision rules. Fixed in the plan before any result. For H2b, $P < .05$ with a positive difference was to be reported as "early renewers' regions spent more relative to their current benchmarks, evidence for the reset motive beyond past savings," $P < .05$ with a negative difference as evidence against it, and otherwise as no detectable difference. H2a used the same wording; H3 predictors were supported, contradicted or undetermined by their direct-simulation *P* values; and H4 was reported as no detectable difference, grew less or grew more, read as an effect only if the equivalence condition held. No test here could rule the sharing-rate motive in or out.

Conventional methods and robustness. Reported beside the registered methods and not used for decisions: Welch *t* tests,[45] ordinary least squares with HC2 errors[46] and maximum-likelihood logistic regression in statsmodels,[47] and 2-way fixed effects with an event study and errors clustered by ACO for H4; and the registered robustness checks (medians, single cohort, adjusted regressions, complete cases, unadjusted county spending, no measurement-error model). Variables were built in dbt[48] on DuckDB;[49] the environment was locked with uv;[50] MCMC diagnostics used ArviZ;[51] every refit had its own seed from a fixed NumPy seed sequence.[52]

Deviations. All 7 analysis stages ran with code identical to the registered commit. After the last stage the lockfile took security updates to 2 development dependencies; no result was produced with it, and it is logged as the one deviation. One sentence of the plan's background is imprecise: it describes ENHANCED as sharing 75% of savings and losses, whereas it shares 75% of savings and up to 75% of losses.

Ethics

The study used only publicly available, aggregate data on organizations and counties published by CMS. It involved no individual-level data, no interaction with people and no identifiable private information, so it was not human participants research, and ethics committee or institutional review board approval and informed consent were not required.

Results

Validation and sample

The rebuilt benchmark passed validation (eTable 1 in Supplement 1). For 101 of the 111 eligible regular renewers, it had a Spearman correlation of 0.94 with the benchmark CMS published and a median absolute percentage error of 1.6%; the slope of actual on rebuilt was 0.944, inside the registered range, so

no calibration was applied. The own-spending part agreed with CMS's benchmark-year spending (Spearman 0.95; error 1.4%), and its error SD set the measurement-error SD at 4.38 percentage points. That SD exceeds the own-spending component's spread among stayers (SD 3.3), so at the level of one ACO that component is mostly noise; the savings rates, which CMS reports directly, are the firmer evidence of how far ACOs were beating their benchmarks.

Early renewers were more often in the January 1, 2020 cohort and at BASIC's two-sided levels C to E in PY2021 than stayers (Table 1); 16 of the 27 BASIC-track early renewers were at the highest level, E. Of those 27, 26 moved to ENHANCED in their new agreements. The expected-gain variables could not be built for 3 early renewers (10.3%) and 36 stayers (13.0%), both above the plan's 10% line. In every one of these 39 ACOs, CMS had suppressed the count of assigned ESRD beneficiaries in every county of the 2021 assignment file, so the ESRD part of regional spending had no county weights. A diagnostic query run after the results, outside the registered pipeline, showed them to be smaller ACOs (median 6,702 and 8,578 assigned beneficiaries for excluded early renewers and stayers, against 10,757 and 15,649 for those included). One early renewer and 3 stayers had hidden values filled by the registered rules.

Table 1. Characteristics of the 334 ACOs in the 2019 and 2020 agreement cohorts, by group

Characteristic	Early renewers	Stayers	Left after PY2021
ACOs, No.	29	277	28
Agreement start, No. (%)			
July 1, 2019	11 (38)	159 (57)	19 (68)
January 1, 2020	18 (62)	118 (43)	9 (32)
Track in PY2021, No. (%)			
BASIC levels A-B (one-sided)	7 (24)	133 (48)	23 (82)
BASIC levels C-E (two-sided)	20 (69)	76 (27)	4 (14)
ENHANCED	2 (7)	68 (25)	1 (4)
Assigned beneficiaries, PY2020, mean (SD), %			
Aged 85 y or older	10.9 (3.7)	12.4 (4.1)	13.3 (5.8)
Female	56.8 (1.9)	57.1 (2.3)	57.3 (2.5)
Savings rate, PY2020, pp			
Mean (SD)	6.4 (3.6)	4.6 (3.9)	2.7 (5.5)
Median	5.8	4.3	1.5
Regional pull, pp of benchmark			
ACOs with the variable, No.	26	241	20
Mean (SD)	5.0 (5.0)	1.8 (4.6)	4.8 (6.7)
Median	3.5	1.4	2.0

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Characteristic	Early renewers	Stayers	Left after PY2021
Own-spending component, pp of benchmark			
ACOs with the variable, No.	26	241	20
Mean (SD)	−5.3 (4.4)	−3.2 (3.3)	−1.8 (4.6)
Median	−4.2	−2.9	−1.0
Regional component, pp of benchmark			
ACOs with the variable, No.	26	241	20
Mean (SD)	3.7 (1.6)	2.2 (1.8)	2.0 (2.3)
Median	3.8	2.1	1.3

Abbreviations: ACO, accountable care organization; pp, percentage points; PY, performance year. Early renewers ended their agreements and began new ones on January 1, 2022; stayers remained in their agreements; leavers left the program after PY2021. The savings rate is CMS's reported savings as a share of the benchmark (positive when spending was below the benchmark). The expected-gain variables could not be built for ACOs whose county counts of assigned beneficiaries with end-stage renal disease were all suppressed (Methods). Leavers were not compared in any registered test.

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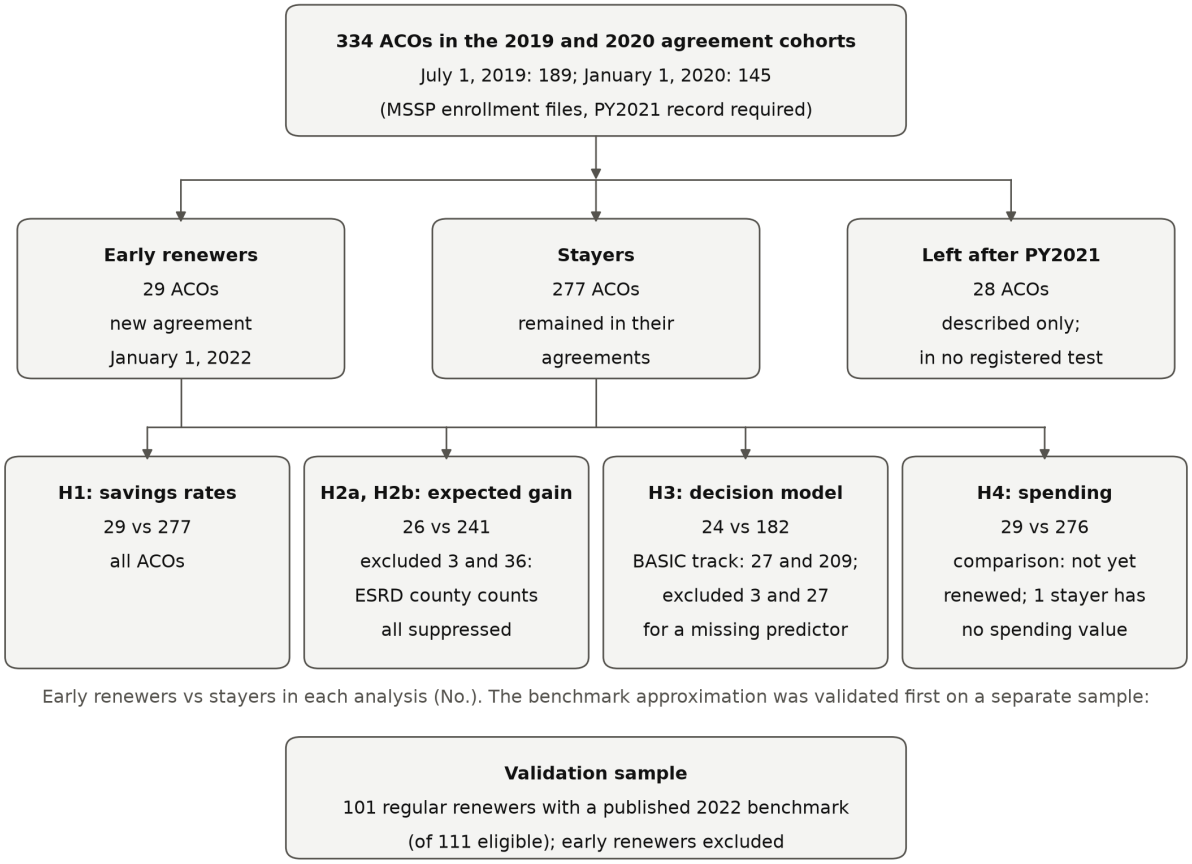


Figure 1. Flow of ACOs into each analysis. Counts are early renewers vs stayers. ACO, accountable care organization; ESRD, end-stage renal disease; PY, performance year.

Who renewed early

Early renewers' regions spent 3.4 percentage points of benchmark more than stayers' regions, relative to their current benchmarks (95% CrI, 1.7 to 5.4; $P < .001$), above the MDE of 2.5 (Table 2 and Figure 2). By the registered rule, H2b was supported: early renewers' regions spent more relative to their current benchmarks, evidence for the reset motive beyond past savings. At the median PY2021 benchmark of the study's ACOs, $11,312 per beneficiary per year, 3.4 points is about $385.

The own-spending component pointed the other way (−2.1 points; 95% CrI, −3.8 to −0.5; $P = .003$): early renewers' recent spending was lower relative to their benchmarks, contrary to the reset motive by the rule's wording. Early renewers also had higher savings rates in PY2020 (1.9 points; 95% CrI, 0.7 to 3.3) and PY2021 (1.6 points; 95% CrI, 0.2 to 3.1). The two parts of the gain offset: the regional component was larger for early renewers (1.6 points; 95% CrI, 0.9 to 2.2), the own-spending component smaller, and the

total expected gain did not differ (−0.5 points; 95% CrI, −2.0 to 0.8; *P* = .33). In unweighted means the own-spending and regional parts summed to −5.3 + 3.7 = −1.6 points for early renewers and −3.2 + 2.2 = −1.0 for stayers: by the approximation, a reset would on average have lowered both groups' benchmarks.

Table 2. Early renewers compared with stayers: regional pull (H2b), own-spending component (H2a), savings rates (H1) and the expected gain

Measure (role)	Early / stayers, No.	Difference (95% CrI)	*P*	MDE	Verdict
H2b, regional pull (primary)	**26 / 241**	**+3.4 (1.7 to 5.4)**	**<.001**	**2.5**	**Supported**
H2a, own-spending component (secondary)	26 / 241	−2.1 (−3.8 to −0.5)	.003	1.8	Contradicted
H1, savings rate, PY2020 (descriptive)	29 / 277	+1.9 (0.7 to 3.3)	.01	2.7	
H1, savings rate, PY2021 (descriptive)	29 / 277	+1.6 (0.2 to 3.1)	.03	2.7	
Total expected gain (descriptive)	26 / 241	−0.5 (−2.0 to 0.8)	.33		
Regional component (descriptive)	26 / 241	+1.6 (0.9 to 2.2)	<.001		

Abbreviations: CrI, credible interval; MDE, minimum detectable effect at 80% power, two-sided α = .05, computed for 29 vs 277 ACOs; pp, percentage points; PY, performance year. Differences are early renewers minus stayers within agreement cohort, averaged with weights proportional to early renewers per cohort. Posterior means and 95% equal-tailed CrIs are from a Bayesian bootstrap (10,000 draws); P values are from stratified permutation (10,000 shuffles within cohort). Expected-gain variables are in percentage points of the current benchmark; savings rates in percentage points. Verdicts follow the registered decision rules; H1 and the two descriptive rows carry none.

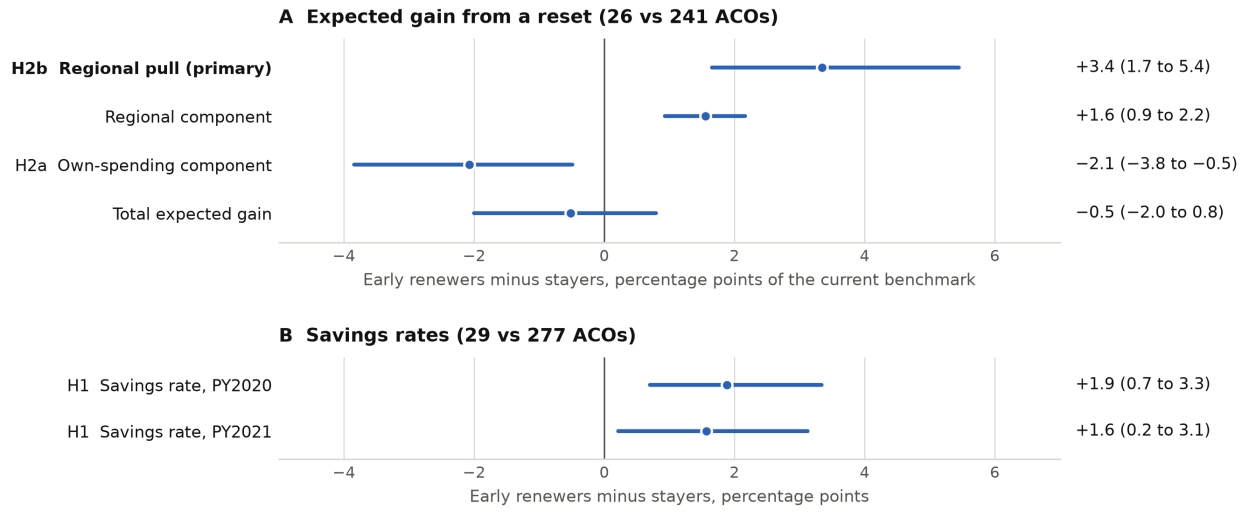


Figure 2. Early renewers minus stayers. A, Expected-gain variables, in percentage points of the current benchmark. B, Savings rates, in percentage points. Points are posterior means and lines 95% credible intervals from the Bayesian bootstrap.

Decision model

Among 206 BASIC-track ACOs (24 early renewers), early renewal was associated with a higher regional pull (log-odds difference per 2 SD, 1.34; 95% CrI, 0.44 to 2.30; direct-simulation *P* = .002) and a higher BASIC level (1.34; 95% CrI, 0.38 to 2.31; *P* = .003); both predictions were supported (Table 3 and Figure 3). A log-odds difference of 1.34 multiplies the odds of renewing early by about 3.8. The own-spending component was undetermined (−0.05; 95% CrI, −2.01 to 1.92; *P* = .19); its interval is the prior's 95% range, because its measurement error exceeds its spread and the model could learn almost nothing about it. Without the measurement-error layer it was −0.64 (95% CrI, −1.51 to 0.23). The model converged on its first fit (largest $\hat{R}$ 1.001, no divergences), and the observed 24 early renewers fell inside the posterior predictive 95% range of 14 to 37 (eTable 5 in Supplement 1). The Student-t prior gave the same verdicts: regional pull 1.65 (95% CrI, 0.53 to 2.87) and BASIC level 1.63 (0.53 to 2.85). The power analysis gave detection rates of only 17% (pull) and 14% (BASIC level) at a log-odds difference of 0.5, so estimates that pass the test are likely to overstate the true effects;[53] the directions are the result, and coefficients near 1.3 are probably too large.

Table 3. Decision model of early renewal among BASIC-track ACOs (H3)

Predictor	Posterior mean (95% CrI)	*P*	Verdict	Student-t prior (95% CrI)	Conventional OR (95% CI)
Regional pull	1.34 (0.44 to 2.30)	.002	Supported	1.65 (0.53 to 2.87)	4.01 (1.20 to 13.40)
Own-spending component	−0.05 (−2.01 to 1.92)	.19	Undetermined	−0.56 (−9.98 to 7.36)	0.53 (0.19 to 1.44)
Higher BASIC level (C-E)	1.34 (0.38 to 2.31)	.003	Supported	1.63 (0.53 to 2.85)	4.79 (1.47 to 15.59)
2020-01-01 cohort	0.63 (−0.24 to 1.51)		Adjustment	0.73 (−0.26 to 1.76)	1.97 (0.71 to 5.49)
Log assigned beneficiaries	0.45 (−0.49 to 1.36)		Adjustment	0.66 (−0.43 to 1.77)	2.00 (0.65 to 6.15)
High revenue	−0.53 (−1.69 to 0.59)		Adjustment	−0.55 (−2.10 to 0.85)	0.63 (0.14 to 2.89)
Prospective assignment	−0.77 (−1.79 to 0.19)		Adjustment	−1.01 (−2.27 to 0.15)	0.37 (0.11 to 1.31)
Later agreement	−0.67 (−1.62 to 0.31)		Adjustment	−0.91 (−2.04 to 0.22)	0.38 (0.12 to 1.19)

Abbreviations: CrI, credible interval; OR, odds ratio. Bayesian logistic regression of early renewal among 206 BASIC-track ACOs (24 early renewers; 3 early renewers and 27 stayers excluded for a missing predictor). Continuous predictors are scaled by 2 SD, so coefficients are log-odds differences per 2 SD; the BASIC level compares levels C-E with A-B. Core predictors (bold) are tested by direct simulation under the reduced model (10,000 simulated datasets); adjustment factors are estimates only. The own-spending component enters with a measurement-error model (σ = 4.38 pp). The Student-t column is the registered prior sensitivity analysis; the maximum-likelihood ORs are the conventional counterpart, reported only.

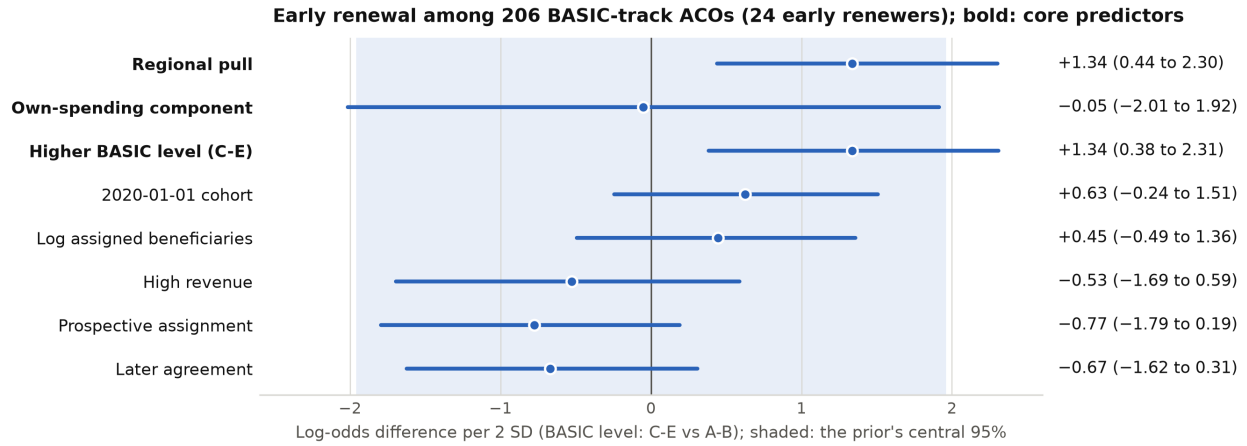


Figure 3. Decision model of early renewal among BASIC-track ACOs (H3). Points are posterior means and lines 95% credible intervals; the shaded band is the Normal(0, 1) prior's central 95%. Bold labels mark the core predictors.

Spending after renewal

Spending growth from PY2020-PY2021 to PY2022-PY2024 did not differ detectably between 29 early renewers and 276 comparison ACOs not yet renewed (−0.4%; 95% CrI, −4.0% to 2.5%; *P* = .78; MDE 3.8 log points) (Table 4 and Figure 4). The pre-trend estimate was 0.7% (95% CrI, −2.3% to 4.9%; *P* = .58), not evidence against parallel trends, but the equivalence condition did not hold: the scaled pre-trend bound, 2.41 years times 0.048, was 0.115 log points, far above the H4 estimate of 0.004. With a single pre-period, that threshold was a property of the design, so for any estimate near zero the condition could only fail. H4 is therefore reported as a difference in spending changes that cannot be told apart from a continuation of a pre-period difference in trends. Under parallel trends, the interval rules out a drop in spending growth of more than about 4% or a rise of more than about 2.5%; the trend-adjusted interval (−12.0% to 2.7%) is too wide to inform.

Attrition from the comparison group was uneven: by PY2024, 115 of 277 stayers (42%) had left the comparison, to exits or later renewals, against 4 of 29 early renewers (14%) (eTable 8 in Supplement 1). The balanced panel of ACOs with all 5 years gave −1.6% (95% CrI, −5.6% to 1.6%; *P* = .30).

Table 4. Spending growth after early renewal (H4), with the pre-trend and robustness estimates

Estimate	Early / comparison, No.	Difference, %	95% CrI	*P*
H4, difference-in-differences	**29 / 276**	**−0.4**	**−4.0 to 2.5**	**.78**
Pre-trend, PY2020 to PY2021	29 / 276	+0.7	−2.3 to 4.9	.58

Estimate	Early / comparison, No.	Difference, %	95% CrI	*P*
Trend-adjusted	29 / 276	−4.0	−12.0 to 2.7	.14
Risk-adjusted outcome	29 / 276	+0.3	−2.0 to 2.5	.85
Balanced panel (all 5 years)	25 / 162	−1.6	−5.6 to 1.6	.30
Propensity-weighted	305 total	−0.5	NA	.67
2020-01-01 cohort only	18 / 118	−2.9	−8.3 to 1.3	.13
Two-way fixed effects (conventional; 95% CI)	306 ACOs	−1.5	−5.4 to 2.4	.44

Abbreviations: CrI, credible interval; NA, not available (the propensity-weighted estimate has a permutation P value but no interval); PY, performance year. The outcome is the change in log per-capita spending from PY2020-PY2021 to PY2022-PY2024, shown as a percent difference in spending growth, early renewers minus ACOs not yet renewed. A stayer's years after it renewed into a 2023 or 2024 agreement are excluded. Equivalence condition: the pre-trend bound scaled to the distance between period midpoints (2.41 years) is 0.115 log points, above the absolute H4 estimate of 0.004, so the condition does not hold and H4 is not read as an effect.

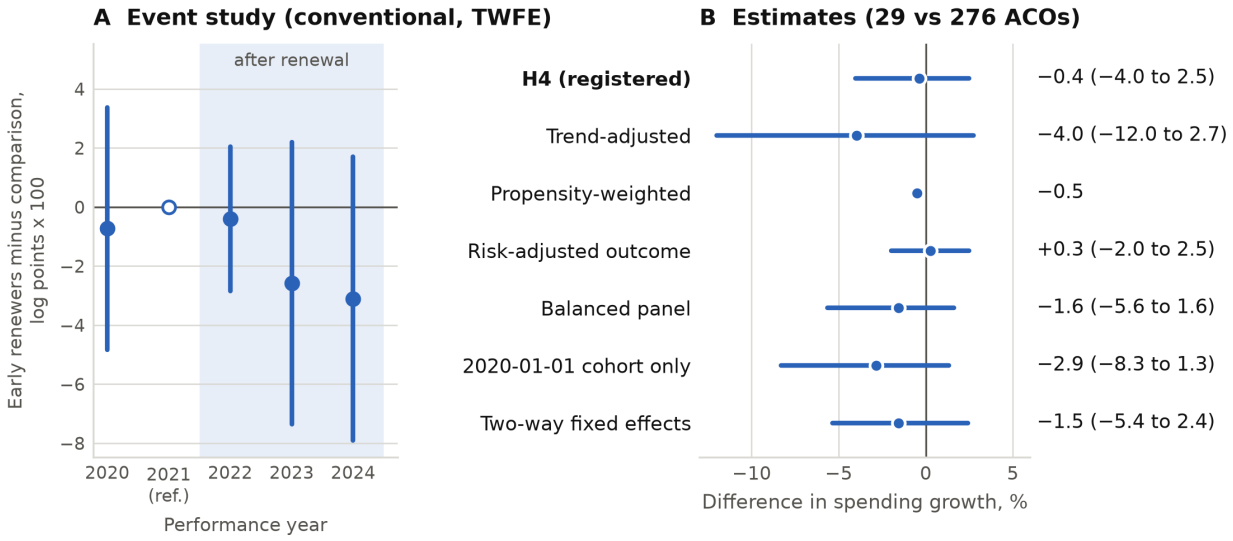


Figure 4. Spending after early renewal. A, Event study from the conventional 2-way fixed-effects model, relative to PY2021, with 95% CIs clustered by ACO. B, The registered H4 estimate and the robustness and conventional estimates, as the percent difference in spending growth with 95% intervals (the propensity-weighted estimate has no interval). TWFE, 2-way fixed effects.

Robustness and conventional methods

Every registered verdict agreed with its conventional counterpart (eTable 7 in Supplement 1). For H2b, ordinary least squares with HC2 errors gave 3.3 points (95% CI, 1.3 to 5.4; *P* = .001), and for H3 the maximum-

likelihood odds ratios were 4.0 (95% CI, 1.2 to 13.4) for the regional pull and 4.8 (95% CI, 1.5 to 15.6) for the BASIC level; 2-way fixed effects gave −1.5% (95% CI, −5.4% to 2.4%) for H4. H2b kept its direction in every robustness check (eTable 3 in Supplement 1), but its size depended on the estimator: the difference in medians was 2.8 points and the cohort-adjusted Bayesian regression 2.1 (95% CrI, 0.5 to 3.8), while without the case-mix adjustment of regional spending the difference was 6.9. In H3 the BASIC-level interval stayed above zero in every check, while the pull interval reached zero with the 2020 cohort alone (1.15; 95% CrI, −0.02 to 2.39) and with unadjusted county spending (0.75; 95% CrI, −0.02 to 1.56) (eTable 4 in Supplement 1). The 28 ACOs that left after PY2021 had the lowest PY2020 savings rates (mean 2.7 points, against 6.4 for early renewers and 4.6 for stayers) (Table 1).

Discussion

In this preregistered study of the 2019 and 2020 MSSP cohorts, the ACOs that ended their agreements early to reset their benchmarks were those whose regions spent more relative to their current benchmarks, and, among BASIC-track ACOs, those already at the higher, two-sided levels. They were also beating their benchmarks by more than ACOs that stayed, so by the study's approximation a reset would not on average have raised their benchmarks: the regional gain and the own-spending loss roughly offset. Spending growth in the 3 years after renewal did not differ detectably from that of ACOs not yet renewed, a result the design could not separate from a continuation of a pre-period difference in trends.

Comparison with earlier work

The primary result extends to a new decision the pattern that ACOs time their participation by the benchmark formula. Participation shifted toward providers whose spending was low relative to their region once regional spending entered benchmarks,[11] and ACOs whose effective benchmarks fell were more likely to leave.[12] Here the choice is when to reset rather than whether to join or leave, and the ACOs that reset early were those with the most room under their regions' spending. The finding that early renewers were savers fits reports that ACOs earning bonuses were more likely to stay in the program,[14,16] but the same success meant that the own-spending part of a reset would have cut their benchmarks, so the simple account that ACOs reset to get a higher benchmark does not describe them on average. That the size of the regional difference doubled without case-mix adjustment is consistent with evidence that comparisons of ACOs with their regions depend heavily on risk adjustment.[10] The concern that rebasing rewards higher spending before a new agreement[9] could be neither confirmed nor excluded here, and the costs to Medicare of early renewal, which earlier work estimated for departures,[18] were not estimated.

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The association with a higher BASIC level bears on the sharing-rate motive at least as much as on the reset motive. In these data renewing early and moving to ENHANCED were one choice: 26 of the 27 BASIC-track early renewers moved to ENHANCED. ACOs already bearing two-sided risk at Level E faced a short step to ENHANCED's 75% sharing rate. A high regional pull is also a plausible reason to want that rate, because an ACO with room under its region's spending expects larger savings to share. No test here can rule either motive in or out.

Implications

For program design, the results suggest that the early-renewal option let ACOs choose when to reset, and that those that used it were positioned to gain from the regional adjustment and the higher sharing rate rather than from a higher benchmark overall. That is selection into an option the program created; it is not evidence that renewing changed spending. The rules have since changed. For agreement periods beginning January 1, 2024 or later, an ACO that saved in its benchmark years can receive a prior savings adjustment, and its benchmark gets the higher of that adjustment and the regional adjustment; negative regional adjustments are partly offset; and the annual update blends in a fixed projected trend.[54] Under these rules a reset costs an ACO that has been beating its benchmark less, so the incentive to renew early, and which ACOs do so, may differ. Two questions remain open: what early renewal cost Medicare, and how much of the choice reflected the sharing rate rather than the benchmark, which a period in which ACOs could move to ENHANCED without resetting would separate.

Limitations

This study has limitations. First, the motives are bundled, and the regional pull, though it has no term for benchmark-year spending, is not free of the ACO's own data: its benchmark was set from earlier spending, and regional spending is adjusted with the ACO's own PY2021 risk scores, so a high pull partly measures how low-cost the ACO had been relative to its region and carries a coding-intensity term.[8] The case-mix adjustment has no published counterpart to validate it. Second, the expected gain is an approximation; validation bounds the error in benchmark levels, but the own-spending component's estimated error exceeds its spread, so H2a and the total gain are weak evidence for any single ACO, and the regional part could not be checked. Third, with 29 early renewers, 24 in the decision model, H3 was underpowered and its supported coefficients are probably overestimates. Fourth, exclusions exceeded the plan's 10% line in both groups, all from suppressed ESRD counts, and the excluded ACOs were smaller, so H2a, H2b and H3 describe the larger ACOs. Fifth, H4 had one COVID-19-affected pre-period, uneven attrition that left the comparison group's later years increasingly selected, possible anticipation (the decision was made during 2021, the last pre-period year and a benchmark year of the new agreement), and possible changes

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in assigned populations through participant lists. Sixth, the base population required a PY2021 enrollment record, and leavers, a third choice, were set aside by comparing early renewers only with stayers. Seventh, the results describe the rules for agreements beginning in 2022. Finally, the analysis reads its table without a fixed row order, so a rerun reproduces deterministic values exactly but resampled and simulated values only within simulation error (for example, H1's simulated power at the MDE was 80.4% as committed and 77.6% in a rerun from a fresh clone), far from changing any verdict.

Conclusions

ACOs that renewed early into 2022 MSSP agreements were those whose regions spent more relative to their benchmarks and those at higher BASIC levels, for whom renewal opened the ENHANCED track. A reset was not, on average, a route to a higher benchmark, and spending growth afterward showed no detectable difference. These are associations under the 2022 rules, measured with public data under a preregistered plan.

Declarations

Data availability: All data are public CMS files.[21,25–27] The list of the 40 files used, with sizes, SHA-256 hashes and URLs, and a script that downloads and verifies them are in the study repository;[24] an archived copy of the snapshot and the results notebook exports are in the OSF project.[28] The committed results files, from which every table and figure here is generated, are in the study repository and in this manuscript's repository (https://github.com/stharrold/mssp-preprint-2026).

Code availability: The registered analysis code (commit a4064f5), its tests and step-by-step instructions to rerun and adapt the analysis are at https://github.com/stharrold/stharrold.github.io (directory analysis/mssp-benchmark-ratchet).[24] A longer account of the study for a general audience was published online.[55]

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Author contributions: STH conceived and designed the study, registered the plan, acquired the data, ran the analysis, interpreted the results, drafted and revised the manuscript, and is accountable for all aspects of the work.

Use of artificial intelligence: The author chose the question, hypotheses and decision rules, approved and registered the plan, ran the analysis, and is responsible for the results and this manuscript. Claude Code (Claude Opus 5.5; Anthropic) drafted the analysis code, the scripts that generate this manuscript's

tables and figures, and parts of the text under the author's direction, and separate instances reviewed the plan, the code and the manuscript. The author reviewed and edited all output and verified the numbers against the results files.

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